Estimating Partition Functions via Quantum Walks

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Immediate Predecessors

- Szegedy, *Spectra of Quantized Walks and a* $\sqrt{\delta\epsilon}$ *rule*, 2004
- Magniez, Nayak, Roland, Santha, *Search via Quantum Walk*, 2006
- Santha, *Quantum Walk Based Search Algorithms*, 2008 (overview article)
- Somma, Boixo, Barnum, Knill, *Quantum Simulations of Classical Annealing Processes*, 2008
- W., Abeyesinghe, *Speed-Up via Quantum Sampling*, 2008

Estimating Parition Functions

- let Ω be a set whose elements x correspond the states of some physical system

- let $E : \Omega \to \mathbb{R}$ denote the energy function, assigning each state x its energy $E(x)$

- given the desired (inverse) temperature β, the task is to estimate the parition function $Z(\beta)$

$$Z(\beta) = \sum_{x \in \Omega} e^{-\beta E(x)}$$

FPRAS

- the problem of computing the partition function exactly is too difficult; it's #P-hard
- $\Rightarrow$ consider fully polynomial randomized approximation schemes
- a FRAPS
  - outputs a random number $\hat{Z}$ satisfying

$$\Pr\Big((1-\epsilon)Z(\beta) \leq \hat{Z} \leq (1+\epsilon)Z(\beta)\Big) \geq 3/4$$

    where $\epsilon \in (0,1)$ determines the desired accuracy
  - runs in time polynomial in $\log(|\Omega|)$ and $1/\epsilon$

Simulated Annealing

- choose a cooling schedule $\beta_0 < \beta_1 < \cdots < \beta_{\ell+1}$ with $\beta_0 = 0$ and $\beta_{\ell+1} = \beta$
- express the desired quantity as a telescoping product

$$Z(\beta) = \frac{Z(\beta_{\ell+1})}{Z(\beta_\ell)} \cdot \frac{Z(\beta_\ell)}{Z(\beta_{\ell-1})} \cdots \frac{Z(\beta_1)}{Z(\beta_0)} \cdot Z(\beta_0)$$

- observe that $Z(\beta_0)$ is trivial since $Z(\beta_0) = |\Omega|$
- $\Rightarrow$ estimate the ratios $\alpha_i := Z(\beta_{i+1})/Z(\beta_i)$

Estimation via Boltzmann Sampling I

- denote by $\pi_i = \big(\pi_i(x) : x \in \Omega\big)$ the Boltzmann distribution at inverse temperature β_i

$$\pi_i(x) = \frac{e^{-\beta_i E(x)}}{Z(\beta_i)}$$

- assume we can sample from π_i
- ensure that each ratio $\alpha_i := Z(\beta_{i+1})/Z(\beta_i)$ is bounded from below by a constant, say $1/2$
- $\Rightarrow$ it suffices to take a small number of samples $\sim \pi_i$ to estimate α_i

Estimation via Boltzmann Sampling II

- let $X_i \sim \pi_i$
- let $\Delta\beta_i = \beta_{i+1} - \beta_i$
- define a new random variable

$$Y_i = e^{-\Delta\beta_i E(X_i)}$$

- the expected value $\mathbf{E}(Y_i)$ is equal to

$$\sum_{x\in\Omega} \pi_i(x) e^{-\Delta\beta_i E(x)} = \sum_{x\in\Omega} \frac{e^{-\beta_i E(x)}}{Z(\beta_i)} e^{(-\beta_{i+1}+\beta_i)E(x)} = \alpha_i$$

- $\Rightarrow Y_i$ is an unbiased estimator for α_i

Estimation via Boltzmann Sampling III

- draw $O(l/\epsilon^2)$ samples of X_i and compute the mean $\bar{Y}_i$
- $\Rightarrow$ the random variable $\bar{Y} = \bar{Y}_0 \bar{Y}_1 \cdots \bar{Y}_\ell$ satisfies

$$\Pr\Big((1-\epsilon)\alpha < \bar{Y} < (1+\epsilon)\alpha\Big) \geq 3/4$$

where $\alpha = \alpha_0 \alpha_1 \cdots \alpha_\ell$

- $\Rightarrow$ the total number of samples is

$$O\Big(\ell^2/\epsilon^2\Big)$$

Sampling with Markov Chains

- in general, we are not able to sample directly from π_i
- $\Rightarrow$ construct a Markov chain P_i such that
  - its stationary distribution is equal to π_i
  - its spectral gap δ is large
    - $\Rightarrow$ we have to simulate

$$\tilde{O}(1/\delta)$$

      steps of P_i to obtain one sample from $\tilde{\pi}_i$
    - $\tilde{\pi}_i$ has to be $O(\epsilon^2/\ell^2)$ close to π_i

Quantum Speed-Up

- classical complexity

$$\tilde{O}\Big(\ell^2/(\delta\epsilon^2)\Big)$$

- quantum complexity

$$\tilde{O}\Big(\ell^2/(\sqrt{\delta}\epsilon)\Big)$$

- $1/\delta \rightarrow 1\sqrt{\delta}$ is due to the quadratic relation between the spectral gaps of P_i and the phase gaps of the corresponding quantum walks $W(P_i)$
- $\epsilon^2 \rightarrow \epsilon$ is due to quantum estimation of expected values of observables

Classical Walk

- P_i is a time-reversible Markov chain

$$P_i = \Big(p_i(x, y) \Big)_{x,y \in \Omega}$$

with stationary distribution

$$\pi_i = \Big(\pi_i(x) \Big)_{x \in \Omega}$$

- π_i is the unique left eigenvector of P_i with eigenvalue 1

$$\pi_i P_i = \pi_i$$

Spectral Gap

- let $1 = \lambda_0 > \lambda_1 > \ldots > \lambda_{|\Omega|-1}$ be the spectrum of P_i
- the spectral gap

$$\delta := 1 - \lambda_1$$

  determines how fast π_i is approached from any initial state

Quantum Update

- a quantum update for P_i is any unitary U_i acting on $\mathbb{C}^{|\Omega|} \otimes \mathbb{C}^{|\Omega|}$ with

$$U_i\Big(|x\rangle \otimes |0\rangle\Big) = |x\rangle \otimes \sum_{y \in \Omega} \sqrt{p_i(x, y)}|y\rangle$$

  for some fixed state $0 \in \Omega$

- classical complexity: the number of times we apply P_i

- quantum complexity: the number of times we apply U_i or $U_i^\dagger$

Quantum Walk

- the quantum walk $W(P_i)$ is realized by applying U_i twice and $U_i^\dagger$ twice (and also some other simple unitaries)
- the unique eigenvector of $W(P_i)$ with eigenvalue 1 is

$$|\pi_i\rangle = \sum_{x\in\Omega} \sqrt{\pi_i(x)}|x\rangle$$

- $|\pi_i\rangle$ is a coherent version of the limiting distribution π_i

Quadratic Relation

- the phase gap Δ of $W(P_i)$ is

$$\min\{|\varphi| : \ e^{i\varphi} \text{ is an eigenvalue of } W(P_i),\ e^{i\varphi} \neq 1\}$$

- the quadratic relation between the phase and spectral gaps

$$\Delta \geq \sqrt{\delta}$$

  is at the heart of quantum speed-ups of many search problems

Primitives

- let

$$\Pi_i := |\pi_i\rangle\langle\pi_i| \quad \text{and} \quad \Pi_i^\perp = I - \Pi_i$$

- we can implement approximate versions of

$$\begin{aligned} R_i &= -\Pi_i + \Pi_i^\perp \\ S_i &= e^{-2\pi j/3}\Pi_i + \Pi_i^\perp \quad \text{and} \quad S_i^\dagger \end{aligned}$$

by applying the quantum walk $W(P_i)$

$$\tilde{O}\Big(\frac{1}{\sqrt{\delta}}\Big)$$

times

Quantum Estimation of the Ratios I

- let A_i be the observable

$$A_i = \sum_{x \in \Omega} e^{-\Delta\beta_i E(x)} |x\rangle\langle x|$$

- we have

$$\langle \pi_i | A_i | \pi_i \rangle = \alpha_i$$

Quantum Estimation of the Ratios II

- let

$$V_i := \sum_{x\in\Omega} |x\rangle\langle x| \otimes \begin{pmatrix} \sqrt{e^{-\Delta\beta_i E(x)}} & \sqrt{1-e^{-\Delta\beta_i E(x)}} \\ -\sqrt{1-e^{-\Delta\beta_i E(x)}} & \sqrt{e^{-\Delta\beta_i E(x)}} \end{pmatrix}$$

- let

$$|\psi_i\rangle = V_i\Big(|\pi_i\rangle \otimes |0\rangle\Big) \quad \text{and} \quad \Gamma := I \otimes |0\rangle\langle 0|$$

- $\Rightarrow$ we have

$$\begin{aligned} \langle\psi_i|\Gamma|\psi_i\rangle &= \langle\pi_i|A_i|\pi_i\rangle \\ &= \alpha_i \end{aligned}$$

Quantum Estimation of the Ratios III

- we apply quantum phase estimation to

$$G = \big(2|\psi_i\rangle\langle\psi_i) - I\big)\big(2\Gamma - I\big) \quad \text{and} \quad |\psi_i\rangle = |\pi_i\rangle|0\rangle$$

- $\Rightarrow$ we get a random variable Q_i such that

$$\Pr\left(\left(1 - \frac{\epsilon}{2(\ell+1)}\right)\alpha_i \leq Q_i \leq \left(1 + \frac{\epsilon}{2(\ell+1)}\right)\alpha_i\right) \geq 1 - \frac{1}{16(\ell+1)}$$

- it suffices to apply the reflection R_i around $|\pi_i\rangle$

$$O\left(\frac{\ell}{\epsilon}\right)$$

times

Quantum Estimation of the Ratios IV

- let Q be the random variable

$$Q = Q_0 Q_1 \cdots Q_\ell$$

- $\Rightarrow$ we have

$$\Pr\Big((1-\epsilon)\alpha \le Q \le (1+\epsilon)\alpha\Big) \ge \frac{15}{16}$$

- it suffices to apply operators from $\{W(P_0), \ldots, W(P_\ell)\}$

$$O\Big(\frac{\ell^2}{\sqrt{\delta\epsilon}}\Big)$$

times

Preparation of Quantum Samples I

- the fact that the ratios α_i are bounded from below by $1/2$ implies

$$|\langle \pi_i | \pi_{i+1} \rangle|^2 \geq \frac{1}{2}$$

- $\Rightarrow$ we can drive $|\pi_i\rangle$ to $|\tilde{\pi}_i\rangle$ by applying operators from the set $\{S_i, S_i^\dagger, S_{i+1}, S_{i+1}^\dagger\}$

$$\tilde{O}(1)$$

  times

- this is based on Grover's $\frac{\pi}{3}$-fixed point search

Preparation of Quantum Samples II

- starting from $|\pi_0\rangle$ (easy because uniform superposition), we can prepare any $|\tilde{\pi}_i\rangle$ by

$$|\pi_0\rangle \rightarrow |\tilde{\pi}_1\rangle \rightarrow \cdots \rightarrow |\tilde{\pi}_i\rangle \rightarrow \cdots \rightarrow |\tilde{\pi}_\ell\rangle$$

- $\Rightarrow$ it suffices to apply the operators from the set $\{S_k, S_k^\dagger : k = 0, \ldots, i\}$ $\tilde{O}(i)$ times

- $\Rightarrow$ the complexity of approximating

$$|\pi_0\rangle \otimes |\pi_1\rangle \otimes \cdots \otimes |\pi_\ell\rangle$$

is

$$\tilde{O}\left(\ell^2\right)$$

Summary

$$|\pi_0\rangle \; — \boxed{\tilde{\alpha}_0} —$$

$$|\pi_0\rangle \rightarrow |\tilde{\pi}_1\rangle \; — \boxed{\tilde{\alpha}_1} —$$

$$|\pi_0\rangle \rightarrow |\tilde{\pi}_1\rangle \rightarrow \cdots \rightarrow |\tilde{\pi}_l\rangle \; — \boxed{\tilde{\alpha}_\ell} —$$

for each $i = 0, \ldots, \ell$,

- the complexity of preparing an approximate version $|\tilde{\pi}_i\rangle$ of $|\pi_i\rangle$ is $\tilde{O}(\ell)$
- the complexity of estimating α_i is $\tilde{O}\left(\frac{\ell}{\sqrt{\delta\epsilon}}\right)$

$\Rightarrow$ overall complexity $\tilde{O}\left(\frac{\ell^2}{\sqrt{\delta\epsilon}}\right)$

Future Research

- quantum speed-up of estimating permanents of matrices with non-negative entries
- quantum speed-up of estimting the volume of a convex polytope?
- quantum speed-up of other classical approximation algorithms?
- estimating the partition functions of quantum Hamiltonians???